

User Manual for 3D-ADE toolkit

This manual helps users understand the utilization of the 3D-ADE toolkit. It is composed of a data-adaptive 3D-deconvolution software (Ada3D) and a 3D-super-resolution uncertainty evaluation software (SQUIRREL3D). Ada3D can be used to deconvolve diffraction-limited or super-resolution 3D fluorescence image stacks without the need to manually calibrate the 3D-PSF. SQUIRREL3D can be used to evaluate the fidelity of the deconvolution results, and can also be used to evaluate the quality of 3D physical super-resolution and credibility of 3D deep-learning super-resolution, under the premise that a low-resolution reference is given.

The code, together with the GUI, is distributed with a MATLAB file (.m), an ImageJ Plugin (.jar), and a direct executable file (.exe). The results given on different platforms are identical. For very large-scale computation, we recommend running the MATLAB GUI or source code.

Besides this manual, you can also watch Supplementary Video 1 provided in the manuscript to see how to install and use this software.

1. Hardware requirements

The code and software can be run with any standard personal computer or workstation.

2. System requirements

The code is tested in a Windows 11 environment.

3. Dependency and installation requirements

- To run the MATLAB source code or GUI

To run or modify the MATLAB code and GUI (.m file), it is required that the computer is installed with full MATLAB 2022b or additional versions.

- To run the ImageJ plugin

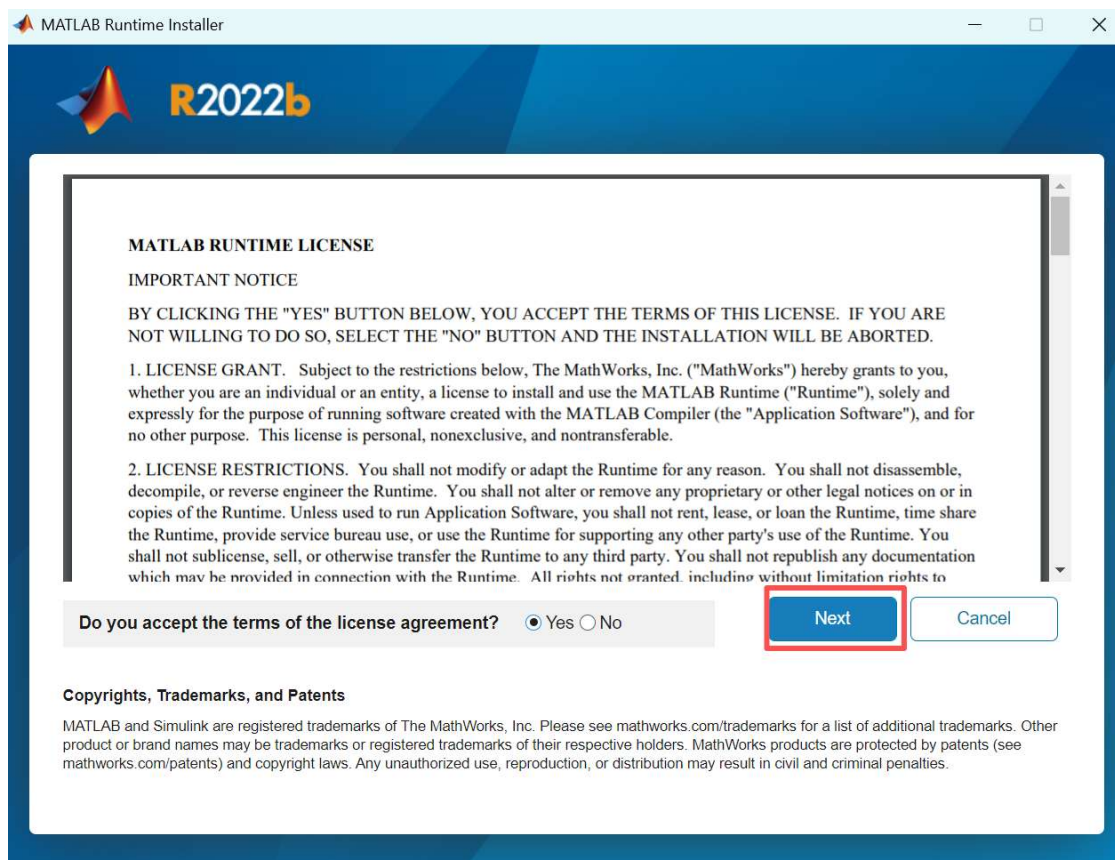
To run the ImageJ plugin (.jar), it is required to install the ImageJ (Fiji) software, and then place the ADE3D.jar file into the Plugin directory of the ImageJ software.

Moreover, it is required to install a runtime matching MATLAB R2022b version:

<https://www.mathworks.com/products/compiler/matlab-runtime.html> (Official link,

freely available) or: <https://doi.org/10.6084/m9.figshare.30730169.v1> , in which we pre-uploaded the runtime file for the convenience of the users. After downloading the zip file, unzip it, double-click setup.exe, then finish the installation using the default settings:

archives	9/15/2025 6:03 PM	File folder	
bin	9/15/2025 6:04 PM	File folder	
cefclient	9/15/2025 6:03 PM	File folder	
extern	9/15/2025 6:00 PM	File folder	
productdata	9/15/2025 6:00 PM	File folder	
remote	9/15/2025 6:00 PM	File folder	
resources	9/15/2025 6:00 PM	File folder	
ui	9/15/2025 6:00 PM	File folder	
utils	9/15/2025 6:04 PM	File folder	
app_uninstaller.zip	9/15/2025 6:04 PM	Compressed (zipped)...	21,817 KB
matlabruntime_installer_input.txt	9/15/2025 6:00 PM	Text Document	2 KB
matlabruntime_license_agreement.pdf	9/15/2025 6:00 PM	Microsoft Edge PDF ...	17 KB
setup.exe	9/15/2025 6:00 PM	Application	506 KB
VersionInfo.xml	9/15/2025 6:00 PM	Microsoft Edge HTM...	1 KB



The requirement of installing a MATLAB Runtime to run the ADE3D code

originates from our observation that a pure Java implementation of the algorithm (with massive 3D-FFT and 3D-Array operations) is too slow.

- To run the executive file

To run the Ada3D software (Ada3D.exe) and SQUIRREL3D software (SQUIRREL3D.exe), it is also required to install a MATLAB runtime.

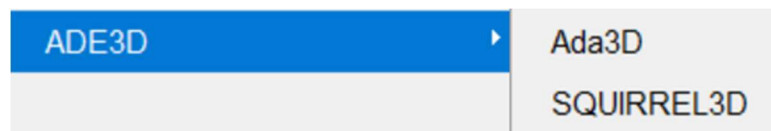
4. How to run the software

- To run the MATLAB source code or GUI

Open the .m file in MATLAB software, and click the run button. Then the GUI will appear.

- To run the ImageJ plugin

After installing the ADE_1.0.0.jar file, open the ImageJ software, and find ADE3D in the “Plugin” menu, click Ada3D or SQUIRREL3D, then the GUI will appear.



- To run the Executive file

Just double-click the .exe file, and then the GUI will appear.

5. Data type requirement

Volumetric image stacks with a .tif file type are required to run the software.

6. Running time

Under the MATLAB testing environment, to perform Ada3D deconvolution on a $1024 \times 1024 \times 40$ 3D-stack, for 40 iterations, it takes ~3 minutes for pure CPU computation (Intel i7-12700, 2.10 GHz), and takes only ~1 minute with GPU acceleration (Nvidia RTX 4080).

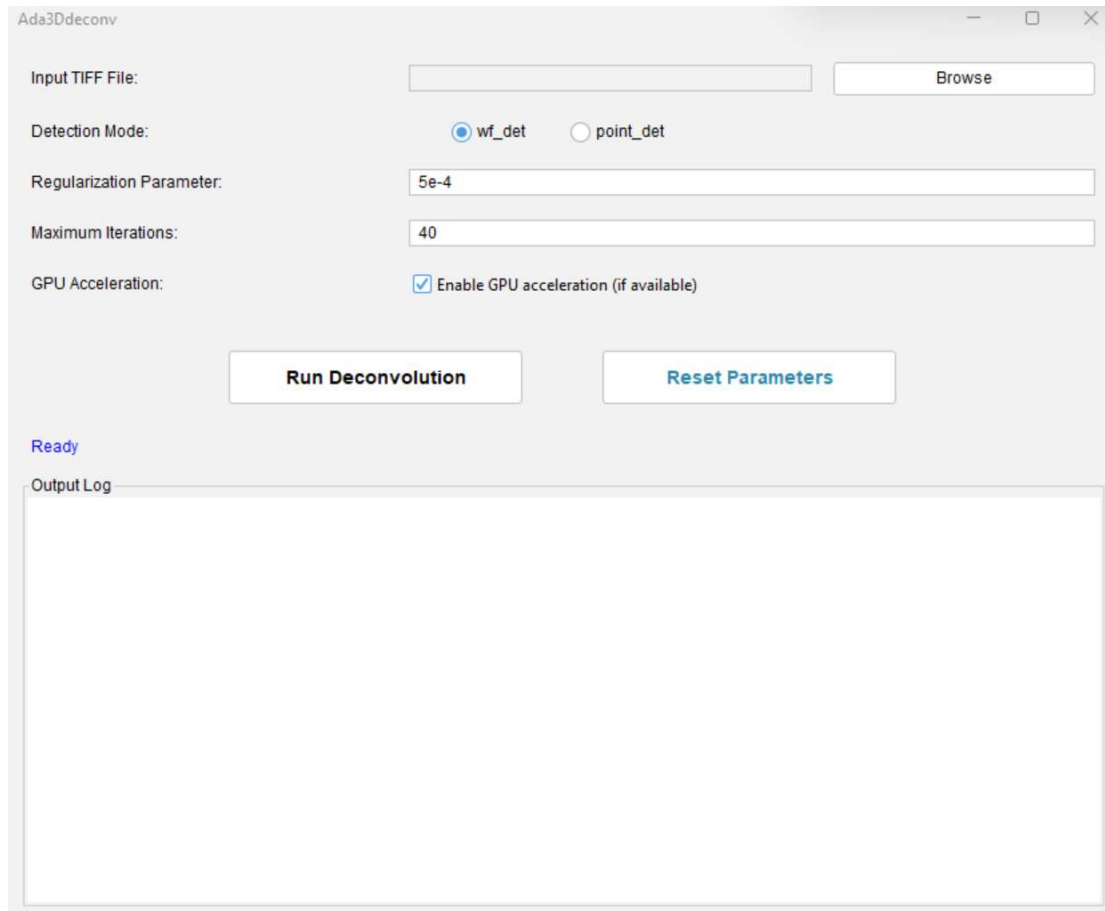
To evaluate the quality of a $1024 \times 1024 \times 40$ 3D-stack with a low-resolution reference of the same size, it takes ~20 seconds for pure CPU computation (Intel i7-12700, 2.10 GHz), and takes only ~6 seconds with GPU acceleration (Nvidia RTX 4080).

7. Demo usage of the ImageJ plugin and exe software

The ImageJ plugin and the exe software share the same GUI and usage method.

7.1 *Ada3D plugin*

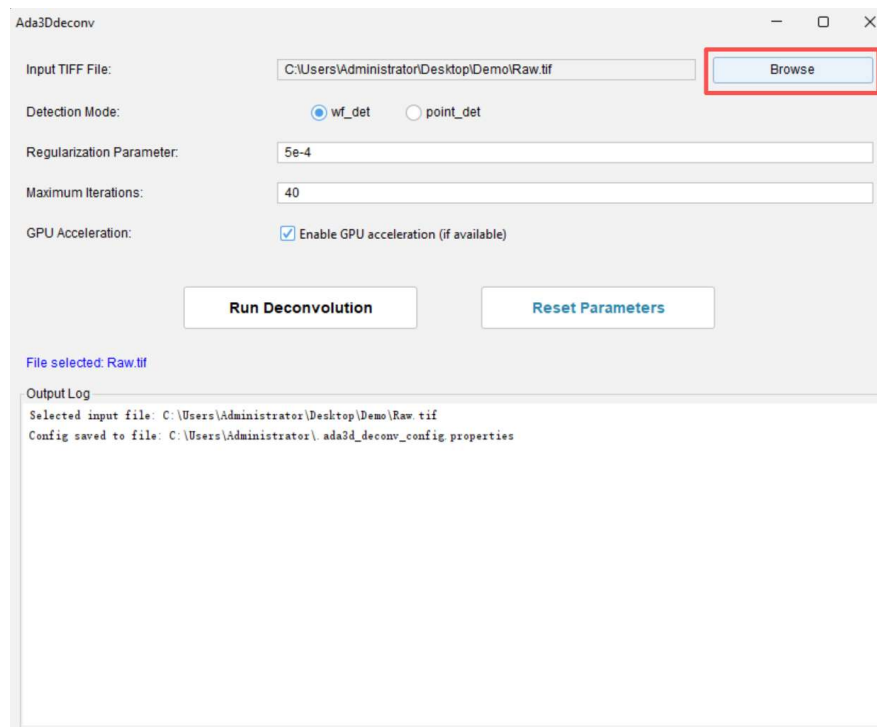
The GUI is devised as follows:



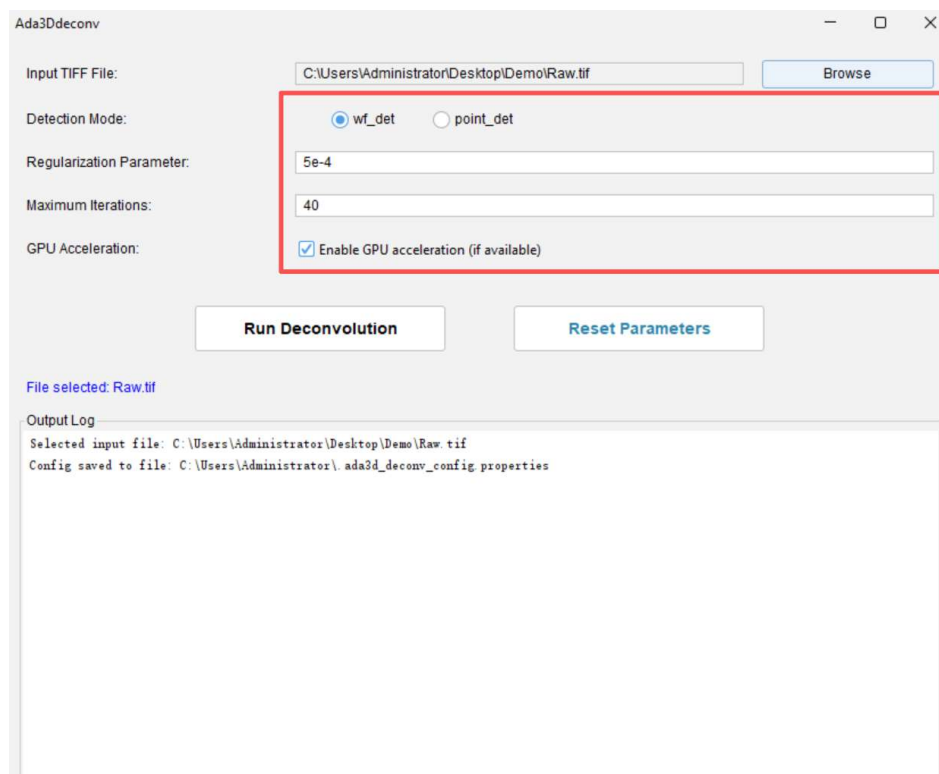
The screenshot shows the 'Ada3Ddeconv' window with the following elements:

- Input TIFF File:** A text input field followed by a 'Browse' button.
- Detection Mode:** Two radio buttons, 'wf_det' (selected) and 'point_det'.
- Regularization Parameter:** A text input field containing '5e-4'.
- Maximum Iterations:** A text input field containing '40'.
- GPU Acceleration:** A checkbox labeled 'Enable GPU acceleration (if available)' which is checked.
- Buttons:** Two buttons at the bottom, 'Run Deconvolution' and 'Reset Parameters'.
- Status:** The word 'Ready' in blue text.
- Output Log:** A large empty text area at the bottom.

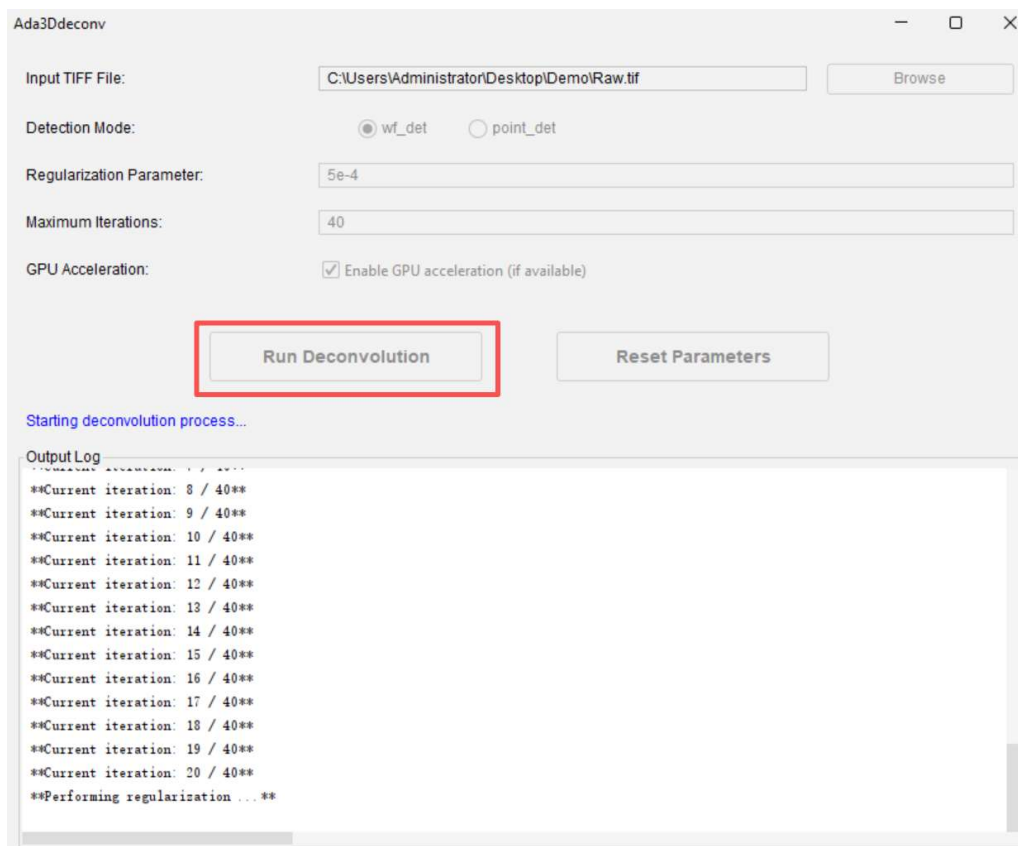
Step 1. Click the “Browse” button and select the data to be processed.



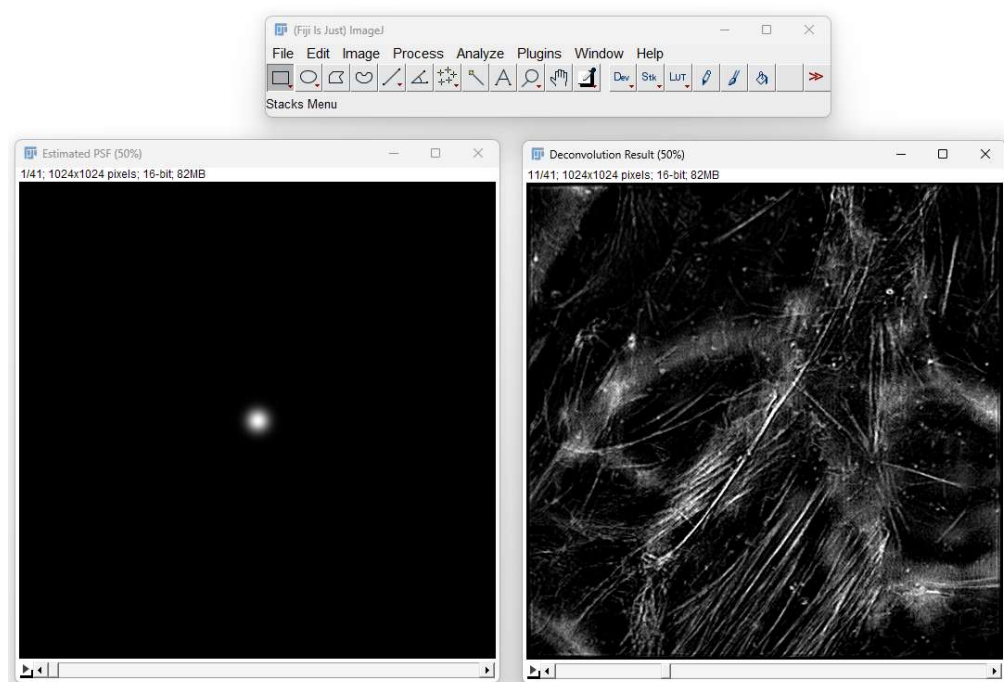
Step 2. Configure the options for 3D-Ada deconvolution. If you do not have a GPU with sufficient memory, please cancel GPU acceleration. Otherwise, it will pop error. Common choice: regularization parameter (1e-5 to 5e-2, it controls noise), maximum iterations (10 to 100).



Step 3. Click the “Run Deconvolution” button. The progress will be displayed in the information box region.



Step 4. Check the results. The deconvolved stack and the estimated 3D-PSF will automatically be presented in the ImageJ software.



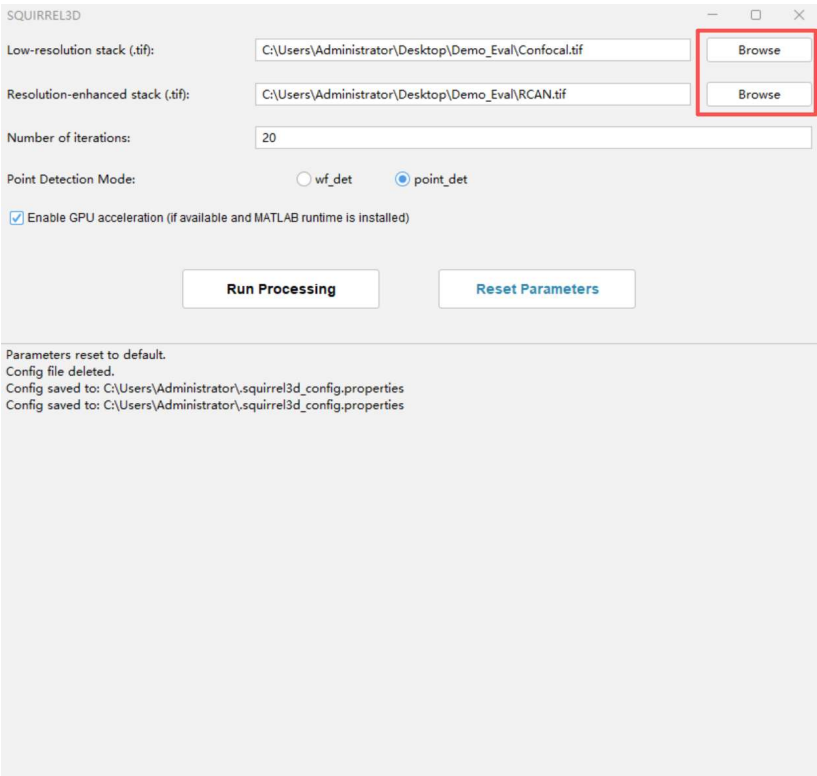
7.2 SQUIRREL3D plugin

The GUI is devised as follows:

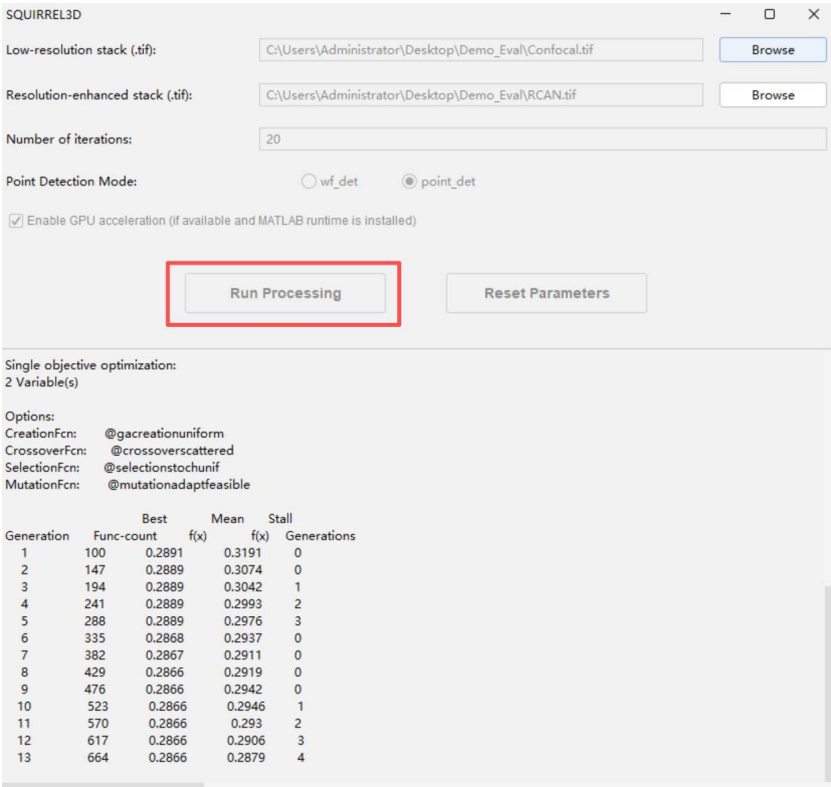
The screenshot shows a MATLAB-style GUI window titled "SQUIRREL3D". It contains the following elements:

- Low-resolution stack (.tif):** A text input field with an empty value and a "Browse" button to its right.
- Resolution-enhanced stack (.tif):** A text input field with an empty value and a "Browse" button to its right.
- Number of iterations:** A text input field containing the value "20".
- Point Detection Mode:** Two radio buttons. The first is labeled "wf_det" and is selected (indicated by a blue dot). The second is labeled "point_det" and is unselected.
- Enable GPU acceleration:** A checkbox that is checked, with the text "Enable GPU acceleration (if available and MATLAB runtime is installed)" next to it.
- Action Buttons:** Two buttons at the bottom: "Run Processing" (white with black text) and "Reset Parameters" (light blue with blue text).
- Status Area:** A large gray area at the bottom containing the text "Parameters reset to default." and "Config file deleted."

Step 1. Input data. Click the two “Browse” buttons and select the low-resolution reference and resolution-enhanced stack. All should be .tif types.



Step 2. Configure the detection mode, then click the “Run Processing” button. The iteration process will be displayed in the blank area of the GUI.



Step 3. Check the results. Usually, a 3D-RSP>0.8 indicates a good SR quality. The results will also be output to the raw data directory and the ImageJ software.

SQUIRREL3D

Low-resolution stack (.tif):

Resolution-enhanced stack (.tif):

Number of iterations:

Point Detection Mode: ☐ wf_det ☒ point_det

☒ Enable GPU acceleration (if available and MATLAB runtime is installed)

Iteration	Points	0.2866	0.2866	0
19	946	0.2866	0.2866	1
20	993	0.2866	0.2866	2

Optimization terminated: maximum number of generations exceeded.
 Best solution: -1.9977 -1.1994
 Fitness value: 0.28655-2.6756e-08i

*****Final evaluation results*****

3D-RSP 0.9037
 3D-RSE 0.0549

File "C:\Users\Administrator\Desktop\Demo_Eval\RCANResolution-scaled_Stack.tif" has been deleted
 File "C:\Users\Administrator\Desktop\Demo_Eval\RCANRSF.tif" has been deleted
 File "C:\Users\Administrator\Desktop\Demo_Eval\RCANerrormap.tif" has been deleted
 MATLAB function executed successfully. Result count: 1
 === Eval3D Core Process Completed Successfully ===
 Releasing resources...
 Released MWArray resource
 Released Eval3D MATLAB instance
 Resource release completed
 === MATLAB Core Process Completed ===
 Opened errormap with SQUIRREL LUT in ImageJ: C:\Users\Administrator\Desktop\Demo_Eval\RCANerrormap.tif

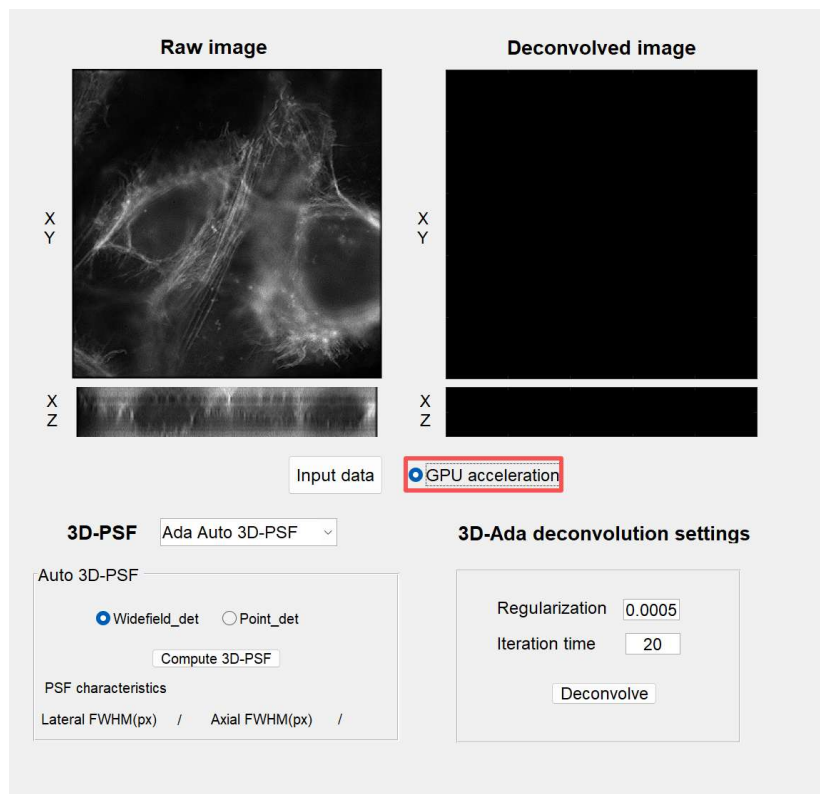
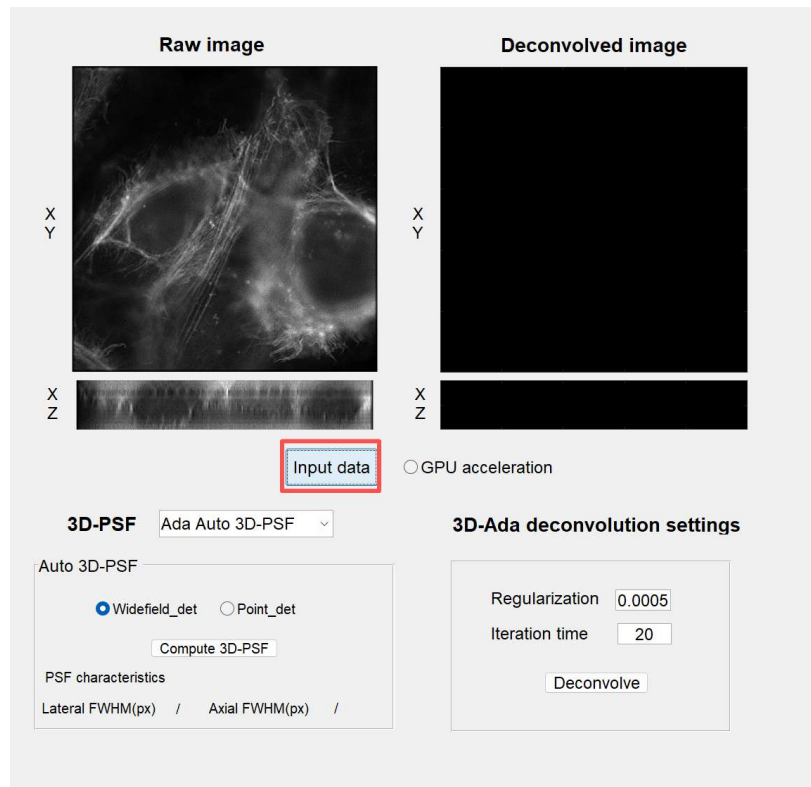
8. Demo usage of the MATLAB software

8.1 Ada3D deconvolution

The GUI is devised as follows:



Step1. Input data. Click the input data button, and load the data. Then the Raw image zone will show the loaded image. Then configure whether to activate the GPU acceleration. If you do not have a GPU with sufficient memory, please cancel GPU acceleration. Otherwise, it will pop error.



Step 2. Configure the 3D-PSF calibration mode. For the automatic 3D-PSF mode, select the detection type of imaging data. Then click the “compute 3D-PSF” button.

Raw image

Deconvolved image

X Y

X Z

Input data

GPU acceleration

3D-PSF

Ada Auto 3D-PSF

Ada Auto 3D-PSF

Manual 3D-PSF

Auto 3D-PSF

Widefield_det

Point_det

Compute 3D-PSF

PSF characteristics

Lateral FWHM(px) / Axial FWHM(px) /

3D-Ada deconvolution settings

Regularization 0.0005

Iteration time 20

Deconvolve

Raw image

Deconvolved image

X Y

X Z

Input data

GPU acceleration

3D-PSF

Ada Auto 3D-PSF

Auto 3D-PSF

Widefield_det

Point_det

Compute 3D-PSF

PSF characteristics

Lateral FWHM(px) 2.223 Axial FWHM(px) 2.627

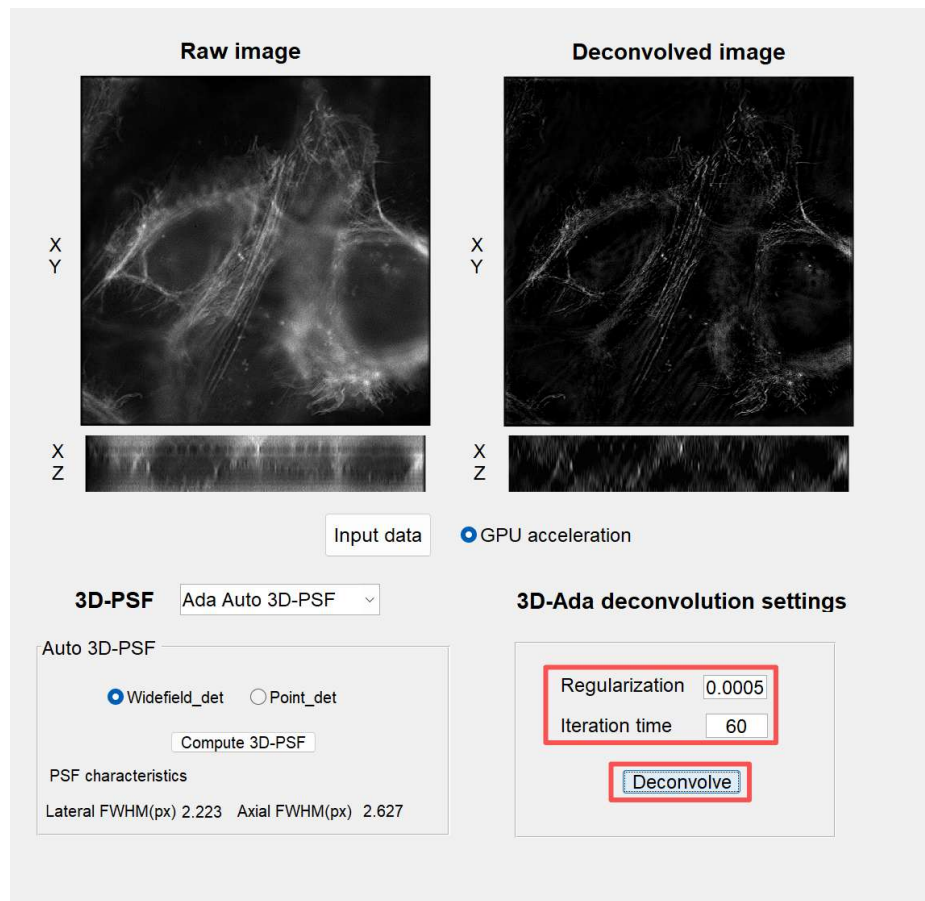
3D-Ada deconvolution settings

Regularization 0.0005

Iteration time 20

Deconvolve

Step 3. Configure the hyperparameters for deconvolution and click the “Deconvolve” button. Common choice: regularization parameter ($1e-5$ to $5e-2$, it controls noise), maximum iterations (10 to 100).

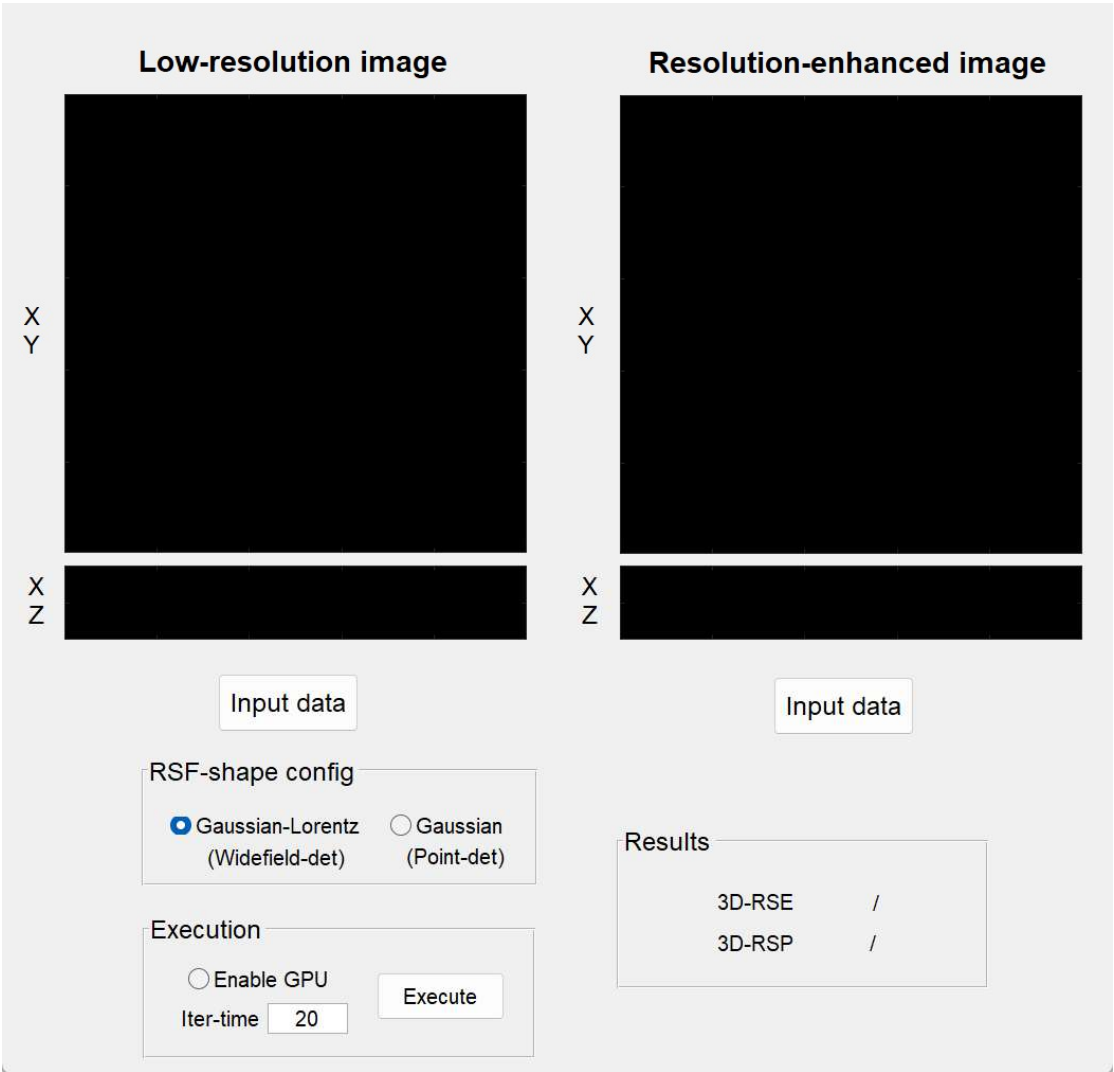


Step 4. Check the deconvolution results

When the deconvolution algorithm is finished, the results would be input into the directory of the original data.

8.2 *SQUIRREL3D* evaluation

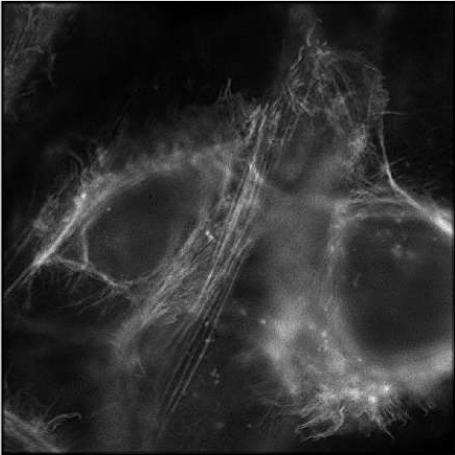
The GUI is devised as follows:



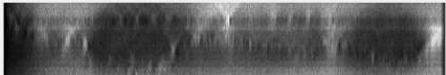
Step 1. Input data. Click the “Input data” button and load the low-resolution reference and resolution-enhanced image, respectively.

Low-resolution image

X
Y



X
Z



Input data

RSF-shape config

☒ Gaussian-Lorentz
(Widefield-det)

☐ Gaussian
(Point-det)

Execution

☐ Enable GPU

Iter-time

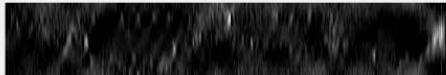
Execute

Resolution-enhanced image

X
Y



X
Z



Input data

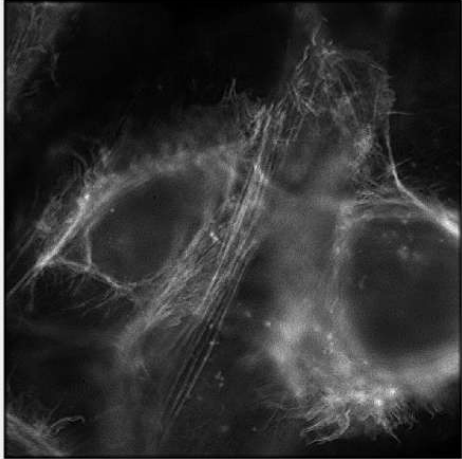
Results

3D-RSE /

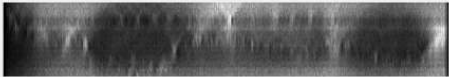
3D-RSP /

Step 2. Configure the RSF-shape. For wide-field detection, we usually choose Gaussian-Lorentz shape. For point detection, we usually choose a Gaussian shape.

Low-resolution image



X
Z



Input data

RSF-shape config

☒ Gaussian-Lorentz
(Widefield-det)

☐ Gaussian
(Point-det)

Execution

☐ Enable GPU

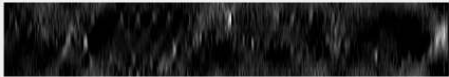
Iter-time

Execute

Resolution-enhanced image



X
Z



Input data

Results

3D-RSE

/

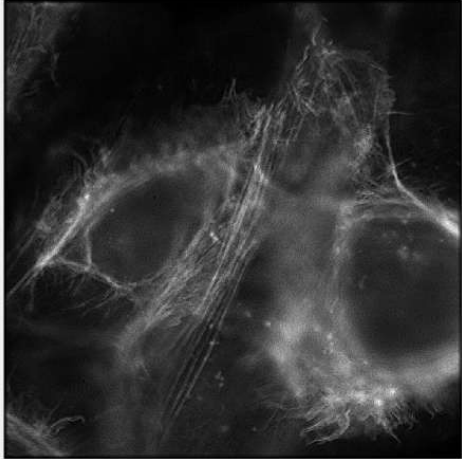
3D-RSP

/

Step 3. Execute the algorithm. Configure the execution choices (usually using the default setting is enough) and click the “Execute” button.

Low-resolution image

X
Y



X
Z

Input data

RSF-shape config

☒ Gaussian-Lorentz
(Widefield-det)

☐ Gaussian
(Point-det)

Execution

☒ Enable GPU

Iter-time

Execute

Resolution-enhanced image

X
Y



X
Z

Input data

Results

3D-RSE

/

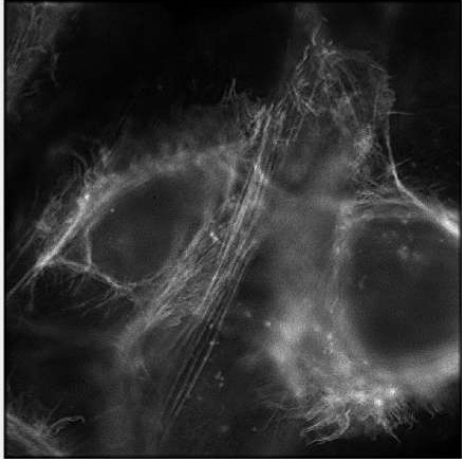
3D-RSP

/

Step 4. Check the results. Usually, a 3D-RSP>0.8 indicates a good SR quality. The detailed results will also be output to the raw data directory.

Low-resolution image

X
Y



X
Z

Input data

RSF-shape config

☒ Gaussian-Lorentz
(Widefield-det)

☐ Gaussian
(Point-det)

Execution

☒ Enable GPU

Iter-time

Execute

Resolution-enhanced image

X
Y



X
Z

Input data

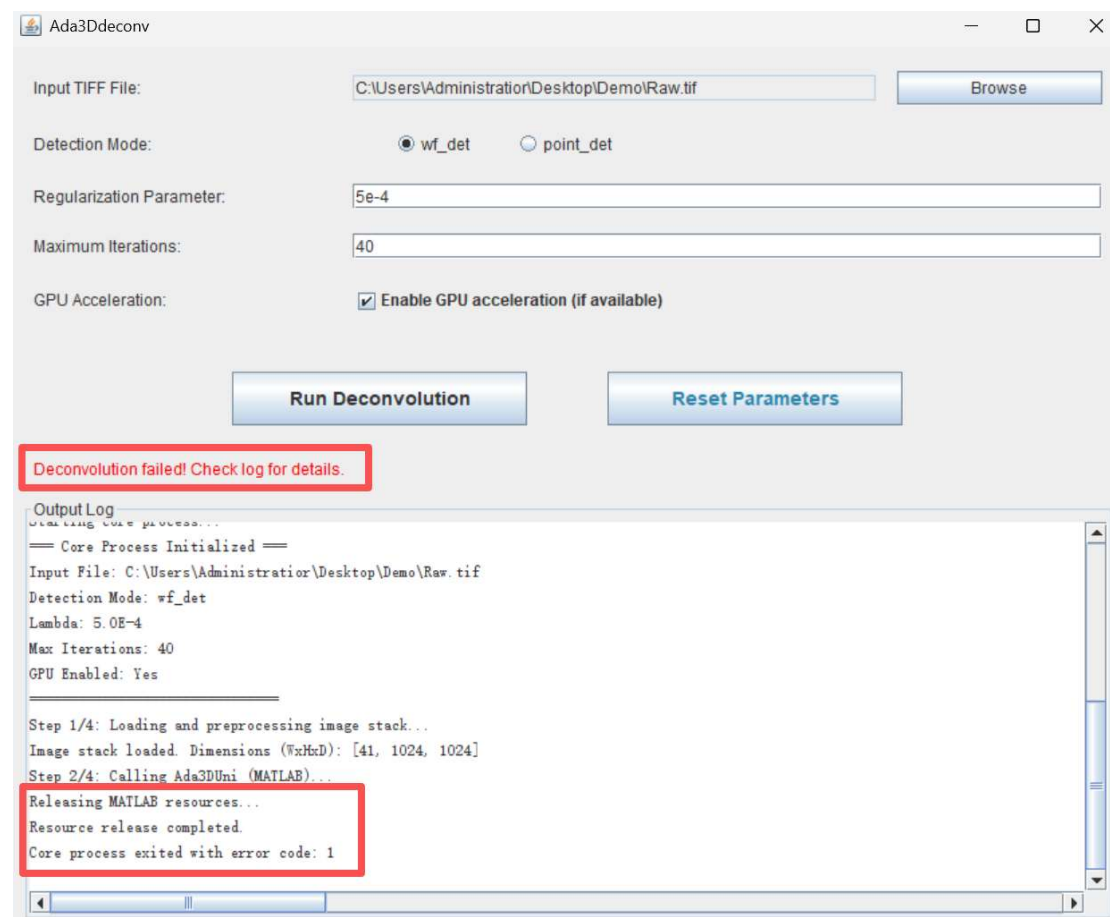
Results

3D-RSE	0.054
3D-RSP	0.883

9 Common Errors

1. MATLAB Runtime version

When using the ImageJ plugin or exe file, if MATLAB runtime 2022b is not installed or a different version is installed, the software will not run successfully with error reminder.



2. GPU out of memory

If you do not have a GPU or the GPU does not have sufficient memory to process current data, the software will pop error like “Out of memory on device”. In this case, you shall cancel the Enable GPU button and use CPU to complete the computation. Moreover, reducing the size of the input image can reduce memory cost.